

Reference Guide

Introduction

Target Device

Contents

1. Overview

1.1 Private LoRa Stack Block Diagram

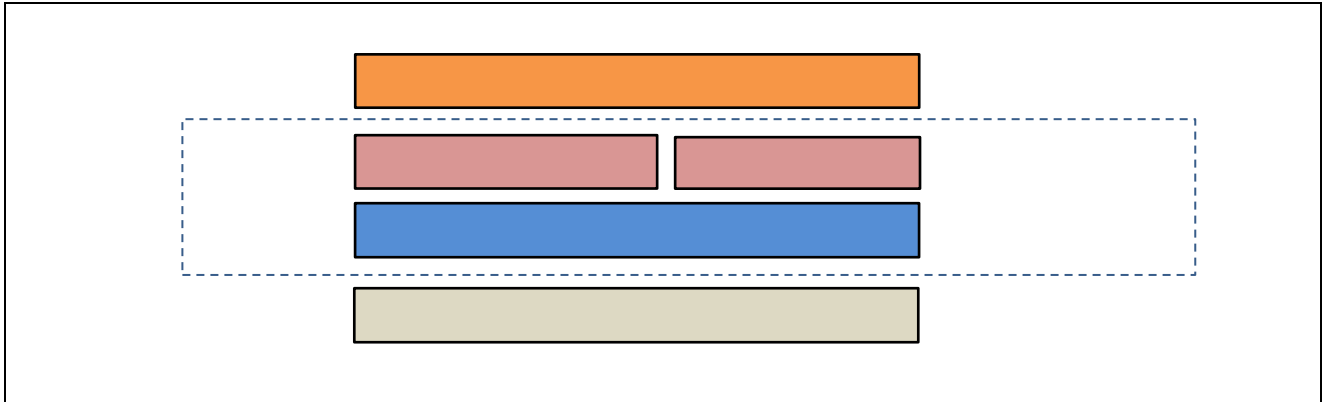


Figure 1. Private LoRa Stack Block Diagram

1.2 Directories (informative)

Table 1. Directories

Directories	Description
src/apps	
src/boards	
src/boards/mcu	
src/prvloramac	
src/radio	
src/peripherals	
src/system	
<ProjectDir>/src/smc_gen	<ProjectDir>

1.3 Resource Usage Example

(package top)\documents\

1.4 Acronyms and Abbreviations

Table 2. Acronyms and Abbreviations

Acronyms	Description

1.5 Related Documentation

	Document No.	Title	Author	Language

2. Private LoRa Interface

2.1 Macros

2.1.1 Stack Settings

Table 3. Macros for Stack Setting

Macro	Description

2.1.2 Configuration

PrivateLoRaConfig.h

Table 4. Macros for Stack Configuration

Macro	Description	
	Type:	
	Type:	
	Type:	
	Type:	
	Type:	

2.2 Enumerations

Table 5. Enumerations

Types	Description

2.2.1 PrvLoRaRegion_t

Table 6. PrvLoRaRegion_t

Enumerator	Description

2.2.2 PrvLoRaStatus_t

Table 7. PrvLoRaStatus_t

Enumerator	Description

2.2.3 PrvLoRaEventInfoStatus_t

Table 8. PrvLoRaEventInfoStatus_t

Enumerator	Description

2.2.4 PrvLoRaIb_t

Table 9. PrvLoRaIb_t

Enumerator	Description
	KeyReq

Table 10. Available Data Rate ID and Channel ID (AS1)

Data Rate ID	Configuration	Indicative physical bit rate	Channel I	Frequency (MHz)	Steps (kHz)

Table 11. Available Data Rate ID and Channel ID (AS2)

Data Rate ID	Configuration	Indicative physical bit rate	Channel I	Frequency (MHz)	Steps (kHz)

Table 12. Available Data Rate ID and Channel ID (AS3)

Data Rate ID	Configuration	Indicative physical bit rate	Channel I	Frequency (MHz)	Steps (kHz)

Table 13. Available Data Rate ID and Channel ID (AS4)

Data Rate ID	Configuration	Indicative physical bit rate	Channel I	Frequency (MHz)	Steps (kHz)

Table 14. Available Data Rate ID and Channel ID (JP)

Data Rate ID	Configuration	Indicative physical bit rate	Channel I	Frequency (MHz)	Steps (kHz)

Table 15. Available Data Rate ID and Channel ID (JP-LDC)

Data Rate ID	Configuration	Indicative physical bit rate	Channel I	Frequency (MHz)	Steps (kHz)

Table 16. Available Data Rate ID and Channel ID (EU)

Data Rate ID	Configuration	Indicative physical bit rate	Channel I	Frequency (MHz)	Steps (kHz)

Table 17. Available Data Rate ID and Channel ID (US)

Data Rate ID	Configuration	Indicative physical bit rate	Channel I	Frequency (MHz)	Steps (kHz)

Table 18. Available Data Rate ID and Channel ID (AU)

Data Rate ID	Configuration	Indicative physical bit rate	Channel I	Frequency (MHz)	Steps (kHz)

Table 19. Available Data Rate ID and Channel ID (IN)

Data Rate ID	Configuration	Indicative physical bit rate	Channel I	Frequency (MHz)	Steps (kHz)

Table 20. Available Data Rate ID and Channel ID (KR)

Data Rate ID	Configuration	Indicative physical bit rate	Channel I	Frequency (MHz)	Steps (kHz)

2.2.5 PrvLoRaMlme_t

Table 21. PrvLoRaMlme_t

Enumerator	Description
	KeyReq
	DevInfoReq
	TxCycleReq

2.2.6 PrvLoRaNotifyId_t

Table 22. PrvLoRaMlme_t

Enumerator	Description

2.3 Structure Types

Table 23. Structure Types

Types	Description

		KeyReq	
			KeyReq
			KeyReq

2.3.2 PrvLoRaTxOptions_t

Table 25. PrvLoRaTxOptions_t (union)

Member type	Member	Description

2.3.3 PrvLoRaMcpsReq_t

Table 26. PrvLoRaMcpsReq_t

Member type	Member	Description

2.3.4 PrvLoRaMcpsCfm_t

Table 27. PrvLoRaMcpsCfm_t

Member type	Member	Description

2.3.5 PrvLoRaMcpsInd_t

Table 28. PrvLoRaMcpsInd_t

Member type	Member	Description

2.3.6 PrvLoRaMlmeReq_t

Table 29. PrvLoRaMlmeReq_t

Member type	Member	Description

2.3.7 PrvLoRaMlmeKeyReq_t

Table 30. PrvLoRaMlmeKeyReq_t

Member type	Member	Description

2.3.12 PrvLoRaMlmeDevInfoCfm_t

Table 35 PrvLoRaMlmeDevInfoCfm_t

Member type	Member	Description

2.3.13 PrvLoRaMlmeTxCycleCfm_t

Table 36 PrvLoRaMlmeTxCycleCfm_t

Member type	Member	Description

2.3.14 PrvLoRaMlmeInd_t

Table 37. PrvLoRaMlmeInd_t

Member type	Member	Description

2.3.15 PrvLoRaMlmeKeyInd_t

Table 38 PrvLoRaMlmeKeyInd_t

Member type	Member	Description

2.3.16 PrvLoRaMlmeTxCycleInd_t

Table 39 PrvLoRaMlmeTxCycleInd_t

Member type	Member	Description
		TxCycleReq

2.4 Private LoRa APIs

Table 40. Private LoRa APIs

function	Description

2.4.1 PrivateLoRaInitialization

PrvLoRaStatus_t PrivateLoRaInitialization(PrvLoRaPrimitives_t *p_primitives, PrvLoRaRegion_t region)	
Parameters	
Return	

2.4.2 PrivateLoRaStart

PrvLoRaStatus_t PrivateLoRaStart(void)	
Parameters	
Return	

2.4.3 PrivateLoRaStop

PrvLoRaStatus_t PrivateLoRaStop(void)	
Return	

2.4.4 PrivateLoRaProcess

void PrivateLoRaProcess(void)	
Parameters	
Return	

2.4.5 PrivateLoRaGetRequest

PrvLoRaStatus_t PrivateLoRaGetRequest(PrvLoRalb_t ibld, PrvLoRalbRequest_t *p_ibGet)	
Parameters	
Return	

2.4.6 PrivateLoRaSetRequest

PrvLoRaStatus_t PrivateLoRaSetRequest(PrvLoRalb_t ibld, PrvLoRalbRequest_t *p_ibSet)	
PrivateLoRaInitialization()	
Parameters	
Return	

2.4.7 PrivateLoRaRegisterRemoteDevice

PrvLoRaStatus_t PrivateLoRaRegisterRemoteDevice(uint8_t *p_remoteMacAddr, uint8_t *p_psk, uint8_t *p_sessionKey uint32_t initFrameCounterTx, uint32_t initFrameCounterRx)	
initFrameCounterTx initFrameCounterRx *p_psk	
Parameters	
Return	

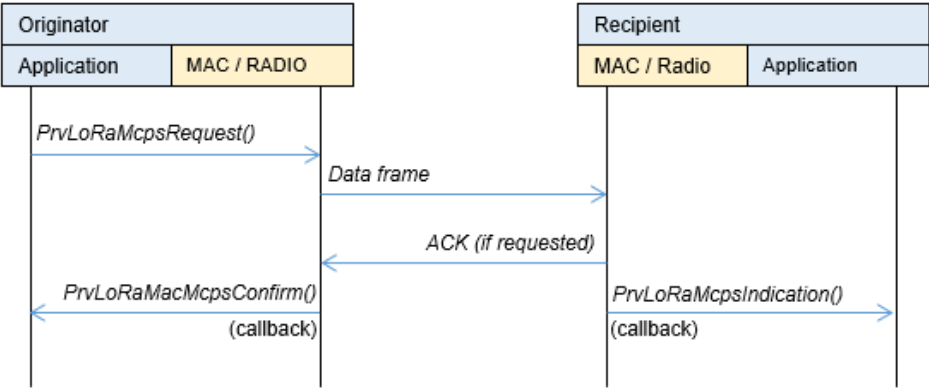
2.4.8 PrivateLoRaUnregisterRemoteDevice

PrvLoRaStatus_t PrivateLoRaUnregisterRemoteDevice(uint8_t *p_remoteMacAddr)	
Parameters	
Return	

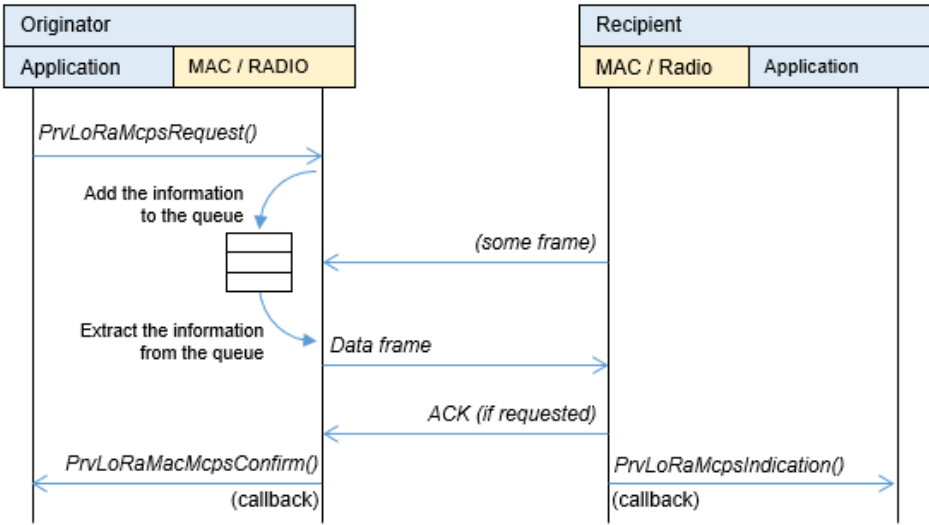
2.4.9 PrivateLoRaMcpsRequest

PrvLoRaStatus_t PrivateLoRaMcpsRequest(PrvLoRaMcpsReq_t *p_mcpsReq)	
PrvLoRaMacMcpsConfirm() PrvLoRaPrimitives_t	
Parameters	
Return	

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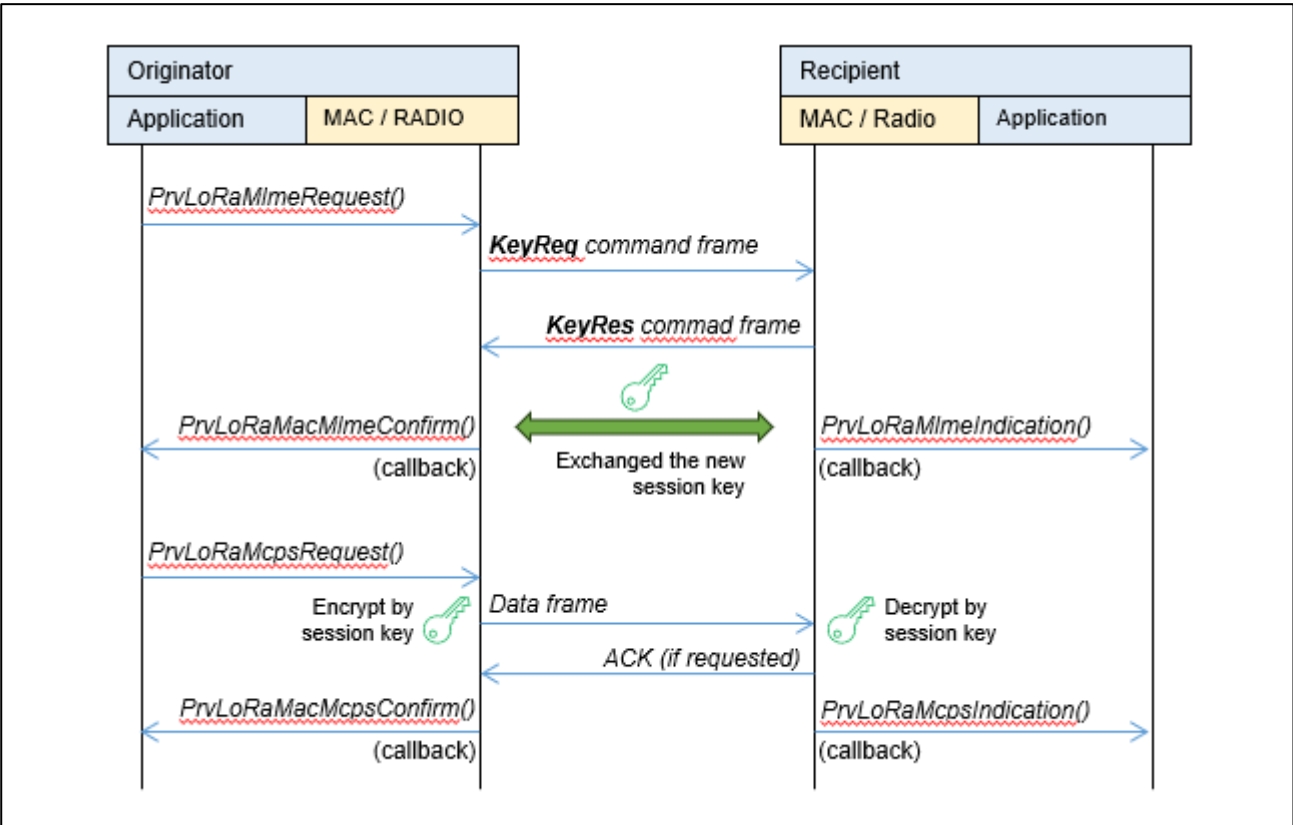
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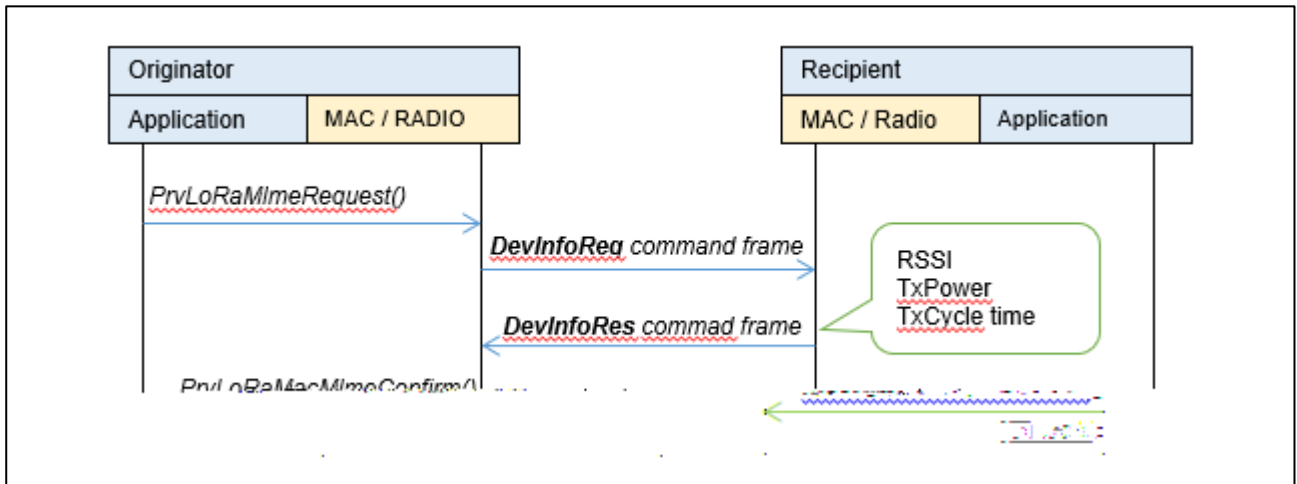
2.4.10 PrivateLoRaMlmeRequest

PrvLoRaStatus_t PrivateLoRaMlmeRequest(PrvLoRaMlmeReq_t *p_mlmeReq)	
PrvLoRaMacMlmeConfirm() PrvLoRaPrimitives_t	
Parameters	
Return	

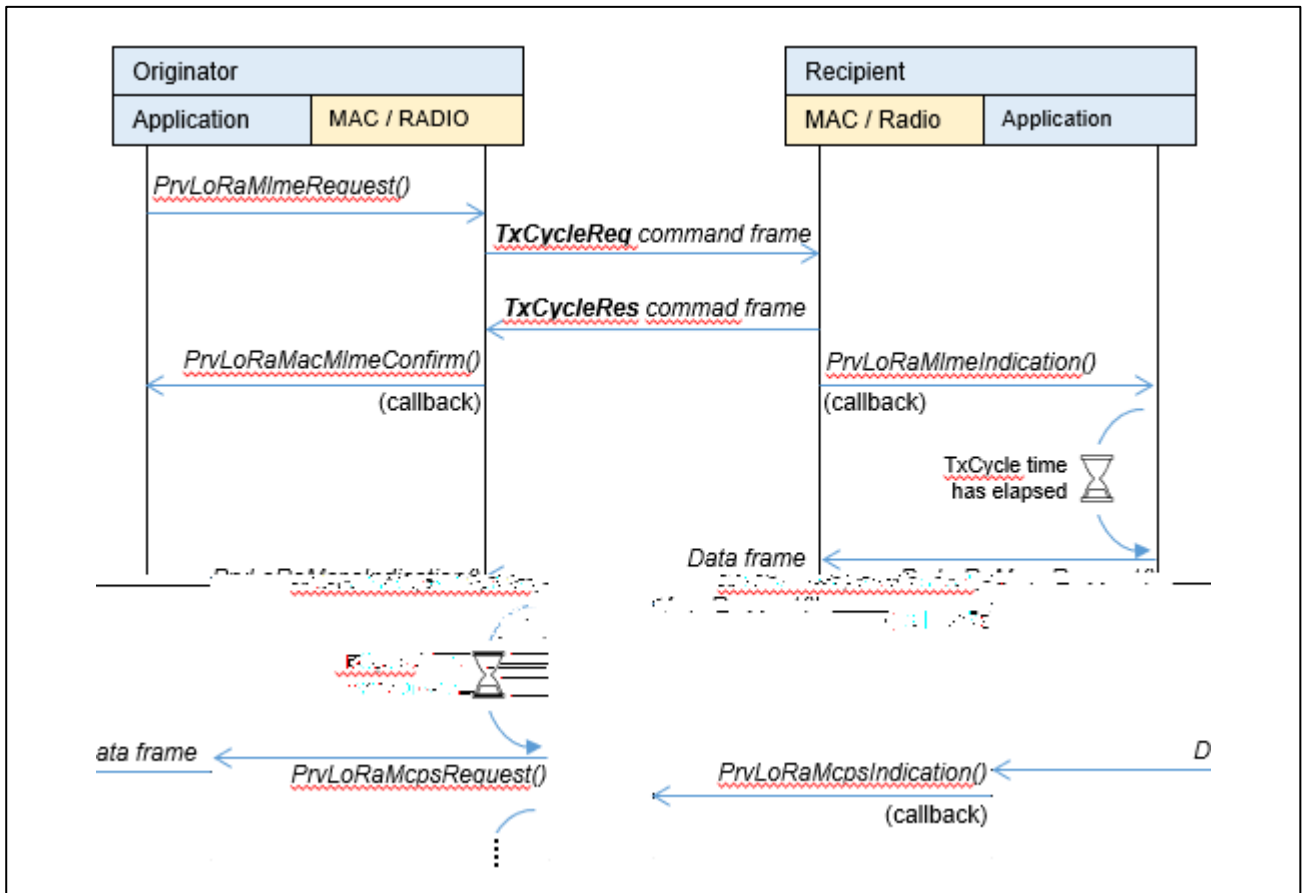
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2.5 Private LoRa Primitive Callback Handler (LoRaMacPrimitives_t)

PrivateLoRaInitialization()

Table 41. PrvLoRaPrimitives_t

Member	Description
	PrivateLoRaMcpsRequest()
	PrivateLoRaMlmeRequest()

2.5.1 PrvLoRaMacMcpsConfirm

void (*PrvLoRaMacMcpsConfirm)(PrvLoRaMcpsCfm_t *p_mcpsCfm)	
PrivateLoRaMcpsRequest()	
Parameters	
Return	

2.5.2 PrvLoRaMacMcpsIndication

void (*PrvLoRaMacMcpsIndication)(PrvLoRaMcpsInd_t *p_mcpsInd)	
Parameters	
Return	

2.5.3 PrvLoRaMacMlmeConfirm

void (*PrvLoRaMacMlmeConfirm)(PrvLoRaMlmeCfm_t *p_MlmeCfm)	
PrivateLoRaMlmeRequest()	
Parameters	
Return	

2.5.4 PrvLoRaMacMlmeIndication

void (*PrvLoRaMacMlmeIndication)(PrvLoRaMlmeInd_t *p_MlmeInd)	
Parameters	
Return	

2.5.5 PrvLoRaMacNotification

void (*PrvLoRaMacMlmeIndication)(PrvLoRaMlmeInd_t *p_MlmeInd)	
Parameters	
Return	

3. Timer

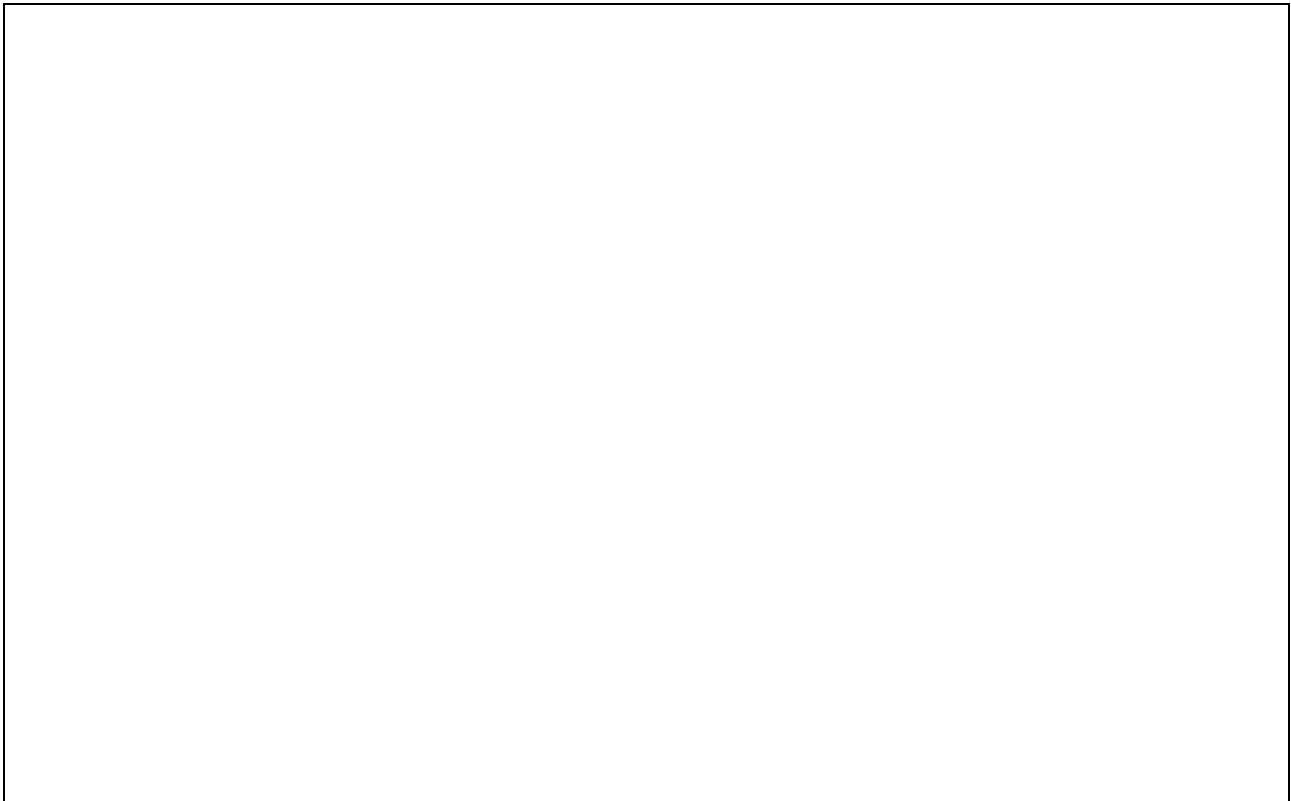
4. Power Saving

4.1 PrivateLoRaSetLowPower

PrvLoRaStatus_t PrivateLoRaSetLowPower(void)	
Parameters	
Return	
Additional explanation	
BoardIsLowPowerAllowed()	
[board.c] bool BoardIsLowPowerAllowed(void)	
BoardIsLowPowerAllowed	

5. Sample Code (Informative)

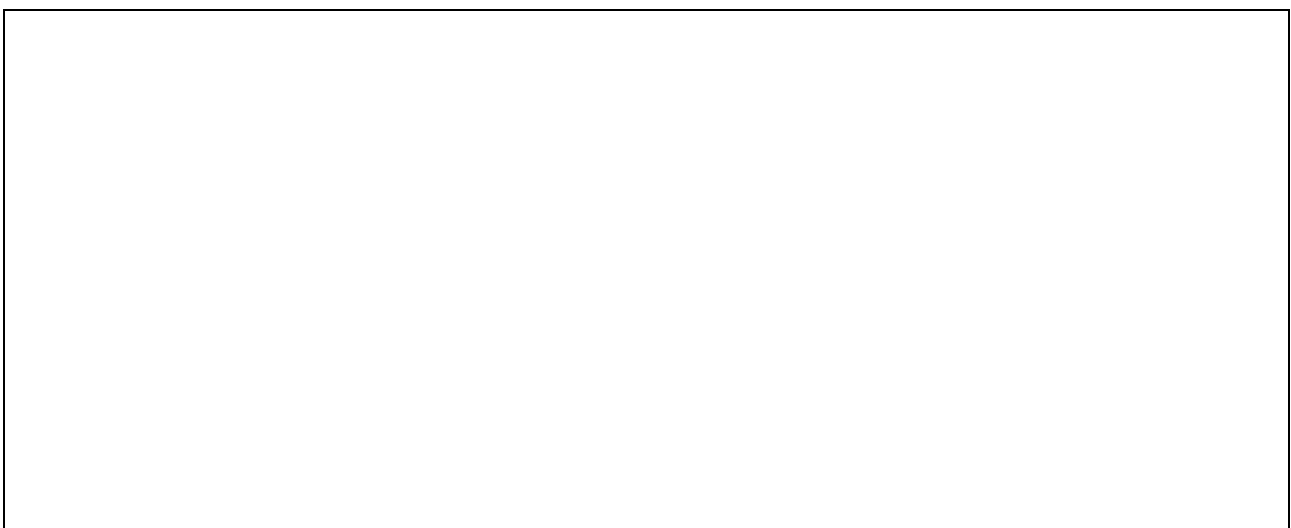
5.1 Initialize and Start the Private LoRa



5.2 Set Private LoRa IB Parameters

data rate ID

MAC address channel ID



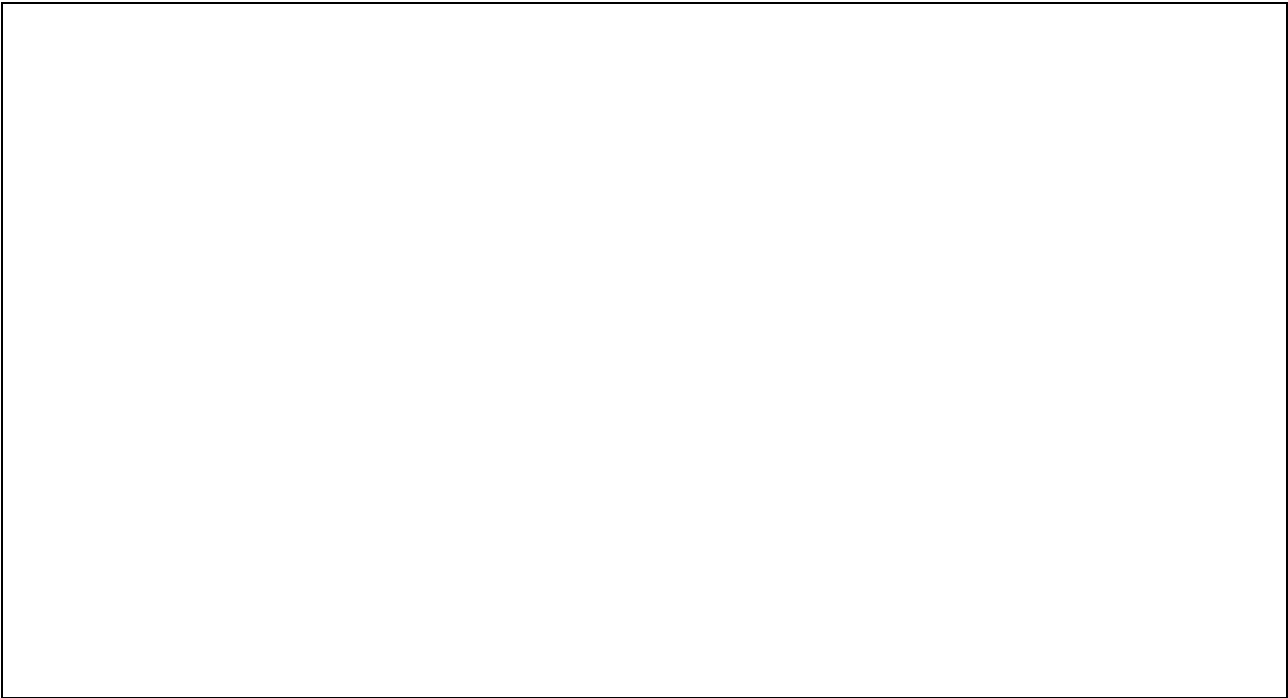
5.3 Get Private LoRa IB Parameters

data rate ID

MAC address channel ID

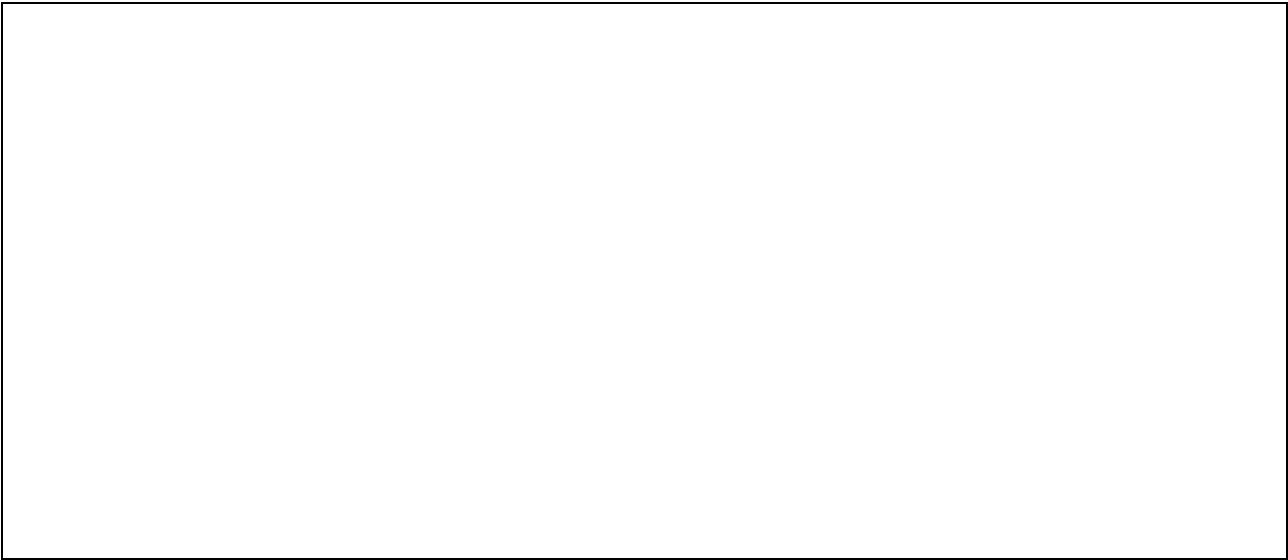
5.4 Remote Device Registration for Communication

5.5 Send Data



5.6 Send Command

KeyReq



Revision History

Rev.	Date	Description	
		Section	Summary

General Precautions in the Handling of Microprocessing Unit and Microcontroller Unit Products

A strong electrical field, when exposed to a CMOS device, can cause destruction of the gate oxide and ultimately degrade the device operation. Steps must be taken to stop the generation of static electricity as much as possible, and quickly dissipate it when it occurs. Environmental control must be adequate. When it is dry, a

Notice